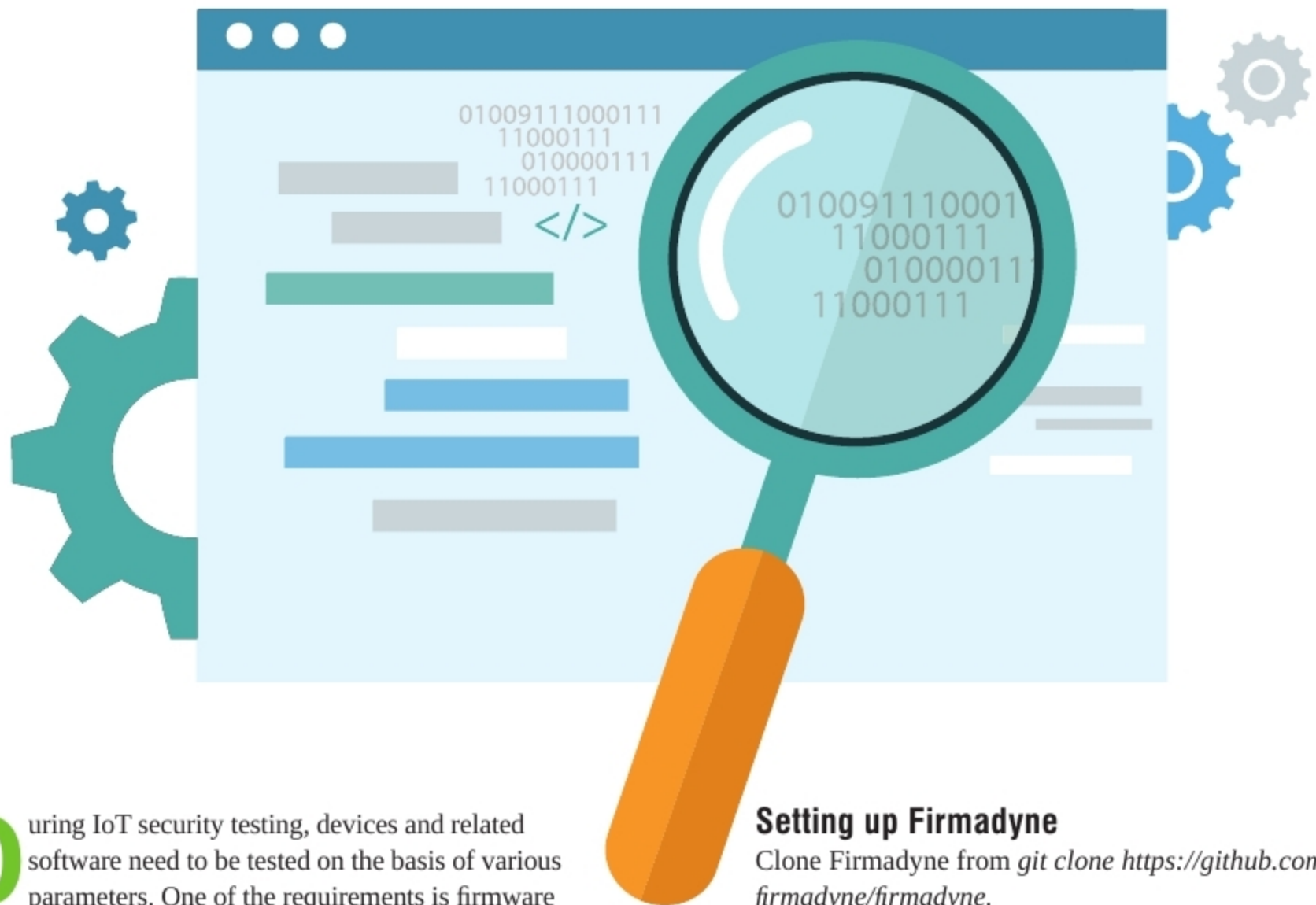


Dynamic Analysis of Firmware Using Firmadyne

Firmadyne is a Linux based system for emulating and analysing firmware. It is used for penetration testing of firmware and can even run exploits. It automatically runs Linux based firmware designed for embedded devices.



During IoT security testing, devices and related software need to be tested on the basis of various parameters. One of the requirements is firmware assessment, which can be carried out with hardware and software emulation. In hardware emulation, the tester has physical access to the device, the firmware is flashed on to it and the tester interacts with it through an interface. The hardware device can be a router, gateway device, etc. In software emulation, the tester may not have access to the device, but the firmware can be emulated through virtualisation, and dynamic testing of the firmware can be carried out.

The advantage of software emulation is that it is cost-effective. There is no need to purchase the device every time the firmware needs to be tested. If the firmware image can be accessed, then it can be mounted by using software emulation techniques and the testing can be done.

Firmadyne is a tool that can be used for dynamic analysis of Linux based embedded firmware. It is built on top of Qemu, an open source machine emulator and virtualiser. It consists of other components such as Binwalk, the PostgreSQL database and binaries, all of which will be discussed in this article.

Setting up Firmadyne

Clone Firmadyne from *git clone <https://github.com/firmadyne/firmadyne>*.

The prerequisites for Firmadyne can be installed with the following commands:

```
apt-get install busybox-static fakeroot git kpartx netcat-
openbsd nmap python-psycpg2 python3-psycpg2 snmp uml-
utilities util-linux vlan
```

During firmware analysis, the first step is to extract the firmware. Generally, firmware is in the *.img*, *.bin* or *.chk* format and needs to be extracted from the image file. This can be done using a tool named Binwalk.

To install Binwalk, use the following command:

```
apt-get install binwalk
```

Firmadyne uses PostgreSQL as the database to store the information and the metadata related to the firmware. The schema for this database can be found at *./firmadyne/database/schema* in the *firmadyne* folder.